

SOFTWARE REQUIREMENTS SPECIFICATION

for

SketchScape – Digital Art Portfolio

Frontend Engineering Project

Software Requirements Specification (SRS) Report

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of SketchScape – Digital Art Portfolio. The document describes the features, system behavior, user interactions, interfaces, and technical requirements needed to develop the application.

1.2 Project Overview

SketchScape is a responsive and interactive web application designed to showcase an artist's sketches, illustrations, and creative works through a visually appealing digital portfolio. Visitors can browse artworks, search and filter them by category, view detailed information about individual artworks, learn about the artist, and contact the artist through a dedicated contact section.

1.3 Objectives

- To provide a professional digital platform for displaying sketches and creative artwork.
- To create an attractive, responsive, and user-friendly interface.
- To allow users to browse, search, and filter artwork easily.
- To provide detailed information about each artwork.
- To demonstrate modern frontend engineering concepts such as reusable components, responsive design, routing, state management, and interactive user interfaces.

1.4 Scope

The scope of SketchScape is limited primarily to frontend development. The application will present artwork using a structured gallery and interactive interface. Core features include a homepage, artwork gallery, category filtering, search functionality, artwork detail views, artist information, creative journey, contact form, theme switching, and responsive design. The first version does not require a complex backend or user authentication system.

2. Overall Description

2.1 Product Perspective

SketchScape is a standalone frontend web application that acts as an online art portfolio. It can be accessed through a modern web browser on desktop, tablet, and mobile devices.

2.2 Target Users

- Visitors interested in viewing artwork and sketches.
- Potential clients interested in contacting the artist for commissions or collaborations.
- Artists who want to present their creative work digitally.
- Students and evaluators reviewing the project as an academic frontend engineering application.

2.3 User Needs

- Users should be able to navigate the website easily.

- Users should be able to browse artworks in an organized gallery.
- Users should be able to search and filter artworks by category.
- Users should be able to open an artwork and view its details.
- Users should be able to learn about the artist and their creative journey.
- Users should be able to submit a message through the contact form.

3. System Requirements

3.1 Hardware Requirements

- Desktop, laptop, tablet, or smartphone capable of running a modern web browser.
- Internet connection for deployment and online access, if the application is hosted.
- Minimum recommended screen resolution: 320 px width for mobile and above.

3.2 Software Requirements

- Operating System: macOS, Windows, Linux, Android, or iOS.
- Modern browser: Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari.
- Development environment: Visual Studio Code or equivalent code editor.
- Node.js and npm for React-based development.

4. Functional Requirements

FR-1: Navigation

The system shall provide a navigation bar that allows users to move between major sections or pages such as Home, Gallery, About, Journey, and Contact.

FR-2: Homepage

The system shall display a visually engaging homepage containing a hero section, introductory text, featured artwork, and a call-to-action button directing users to the gallery.

FR-3: Artwork Gallery

The system shall display artworks in a responsive grid or gallery layout. Each artwork card shall show an image, title, and category.

FR-4: Search Functionality

The system shall allow users to search for artworks using keywords such as artwork titles or categories.

FR-5: Category Filtering

The system shall allow users to filter artworks by categories such as Portraits, Nature, Anime, Architecture, and Abstract.

FR-6: Artwork Details

When a user selects an artwork, the system shall display detailed information including the artwork image, title, category, medium, date of creation, and description.

FR-7: Artist Profile

The system shall provide an About section containing information about the artist, artistic inspiration, creative interests, and preferred artistic mediums.

FR-8: Creative Journey

The system shall display a timeline or journey section showing important milestones in the artist's creative development.

FR-9: Contact Form

The system shall provide a contact form containing fields for name, email, subject, and message. Basic client-side validation shall ensure that required fields are completed and that the email format is valid.

FR-10: Theme Switching

The system may provide a dark/light theme toggle. The selected preference can be stored locally in the browser using Local Storage.

FR-11: Responsive Design

The system shall adapt its layout and content to different screen sizes, including mobile phones, tablets, laptops, and desktop computers.

5. Non-Functional Requirements

5.1 Usability

The interface should be simple, intuitive, and easy to navigate. Buttons, menus, and interactive elements should provide clear visual feedback.

5.2 Performance

The application should load efficiently. Images should be appropriately optimized, and unnecessary resources should be avoided.

5.3 Compatibility

The application should function correctly on current versions of major web browsers and across commonly used device sizes.

5.4 Accessibility

Images should include meaningful alternative text where appropriate. The interface should maintain readable text, sufficient contrast, keyboard-friendly navigation where possible, and clearly identifiable controls.

5.5 Maintainability

The codebase should use a modular component-based structure so that sections and features can be updated or extended easily.

6. External Interface Requirements

6.1 User Interface

The user interface will use a clean, artistic, and modern visual style. Major interface components include a navigation bar, hero section, gallery cards, category filters, search bar, artwork detail view, artist profile, timeline, contact form, and footer.

6.2 Input Interfaces

- Search input for locating artworks.
- Category filter buttons or controls.
- Theme toggle control.
- Contact form fields for user messages.

6.3 Output Interfaces

- Artwork cards and gallery views.
- Filtered and searched artwork results.
- Detailed artwork information.
- Form validation and submission feedback messages.

7. Technology Stack

- HTML5 – semantic structure and content.
- CSS3 – styling, responsive layouts, animations, Flexbox, and Grid.
- JavaScript (ES6+) – interactivity, filtering, search, and client-side logic.
- React.js – component-based frontend development.
- React Router – navigation between application pages or views.
- Local Storage – optional persistence of theme preferences or favorite artworks.
- Git and GitHub – version control and project hosting.
- Vite – optional development and build tool for the React application.

8. System Modules

8.1 Home Module

Displays the main introduction, visual identity of the portfolio, featured artwork, and navigation to other sections.

8.2 Gallery Module

Manages the display of artworks and provides search and category filtering functionality.

8.3 Artwork Details Module

Displays detailed information about a selected artwork and may provide previous/next navigation or a modal/lightbox view.

8.4 Artist Module

Provides information about the artist, creative inspiration, artistic mediums, and personal creative journey.

8.5 Contact Module

Allows visitors to submit inquiries, feedback, or commission requests through a validated contact form.

8.6 Theme Module

Controls dark and light appearance modes and optionally stores the selected preference in Local Storage.

9. Data Requirements

The initial version of SketchScape can store artwork information locally in JavaScript or JSON data files. Each artwork record may contain the following attributes:

- Artwork ID
- Title
- Image path or URL
- Category
- Medium
- Date created
- Description
- Featured status

10. Future Enhancements

- User accounts and artist authentication.
- Backend database for dynamic artwork management.
- Admin dashboard for uploading and editing artwork.
- Artwork favorites synchronized across devices.
- Online commission booking system.
- Social sharing functionality.
- Image zoom and advanced gallery viewing.
- Comments or visitor feedback system.

- Integration with cloud storage or a content management system.

11. Conclusion

SketchScape – Digital Art Portfolio is designed as a creative and responsive web application that combines visual presentation with modern frontend engineering practices. The project provides an organized platform for showcasing artwork while implementing important frontend concepts such as responsive design, reusable components, routing, filtering, search functionality, form validation, local storage, and interactive user interfaces. The proposed system is practical, achievable within the project timeline, and can be expanded with additional features in future versions.